

Fitness Challenge Community

Project Synopsis

1. Introduction / Background

Fitness tracking and group accountability have become central to how people stay active, with wearables and mobile apps making it easier to monitor daily activity. This project develops a social fitness platform where users can log workouts, track progress visually, and compete with others in group challenges — combining habit tracking with community motivation.

2. Problem Statement

- Solo workout tracking lacks motivation and accountability
- Users struggle to visualize physical progress over time without a structured comparison tool
- Fitness data from wearables stays siloed and isn't tied to social or competitive engagement
- Trainers lack an integrated way to assign plans and monetize their programs

3. Objectives

- Build a social fitness app where users log workouts, track progress, and compete in group challenges
- Enable progress photo comparisons with side-by-side before-and-after views
- Create leaderboards ranking participants by challenge points, streak days, or total activity
- Integrate wearables APIs (Fitbit, Apple Health) to auto-sync steps and calorie data

4. Proposed System & User Roles

The system allows users to log workouts, join or create group challenges, sync wearable data automatically, and track visual progress through photo comparisons. It supports the following roles:

- Member – Logs workouts, joins challenges, tracks progress, and views leaderboards
- Challenge Organizer – Creates group challenges with rules, duration, and entry fees
- Trainer – Builds and assigns workout plans, earns payouts from subscriptions/programs
- Admin – Oversees platform activity, payments, and content moderation

5. Basic Workflow

Sign Up → Connect Wearable (optional) → Log Workout / Auto-Sync Activity → Join or Create Challenge → Track Progress on Leaderboard → Upload Progress Photos → Earn Streaks/Badges → View Analytics.

6. Technologies Used

- Frontend – React Native for mobile; React.js for the web dashboard

- Backend – Node.js/Express REST API with scheduled jobs for leaderboard recalculation
- Wearables – Fitbit API and Apple HealthKit integration for automatic activity import
- Media – Cloudinary for progress photo storage with client-side compression
- Database – MongoDB for flexible workout and challenge schemas

7. Advanced Features

- Wearables Sync – OAuth-based connection to Fitbit, Garmin, and Apple Health
- Group Challenges – Multi-participant challenges with join codes and countdown timers
- Real-Time Leaderboards – Ranked boards updating after each logged or synced activity
- Progress Photos – Private gallery with a timeline slider for transformation tracking
- Trainer-Assigned Workout Plans – Progression tracking for assigned programs
- Streak Tracking – Daily activity streaks with milestone badges and reminders
- Nutrition Logging – Calorie/macro tracking with barcode scanning
- Social Feed – Friends' activity feed with likes and comments
- Analytics Dashboard – Personal stats on volume, frequency, PRs, and body metrics
- Offline Mode – Offline workout logging with background sync
- Payments – Stripe-based premium memberships, paid challenge entry fees with prize pools, and automated trainer payouts via Stripe Connect

8. Expected Outcomes

A fully functional, mobile-responsive fitness community application enabling users to log workouts, sync wearable data, compete in group challenges, and visually track progress — while giving trainers a monetizable platform to deliver and manage fitness programs.

9. Conclusion

This project offers an engaging, community-driven fitness solution that blends workout tracking, social competition, and wearable integration into a single platform — benefiting everyday users seeking motivation and trainers seeking a monetizable audience.